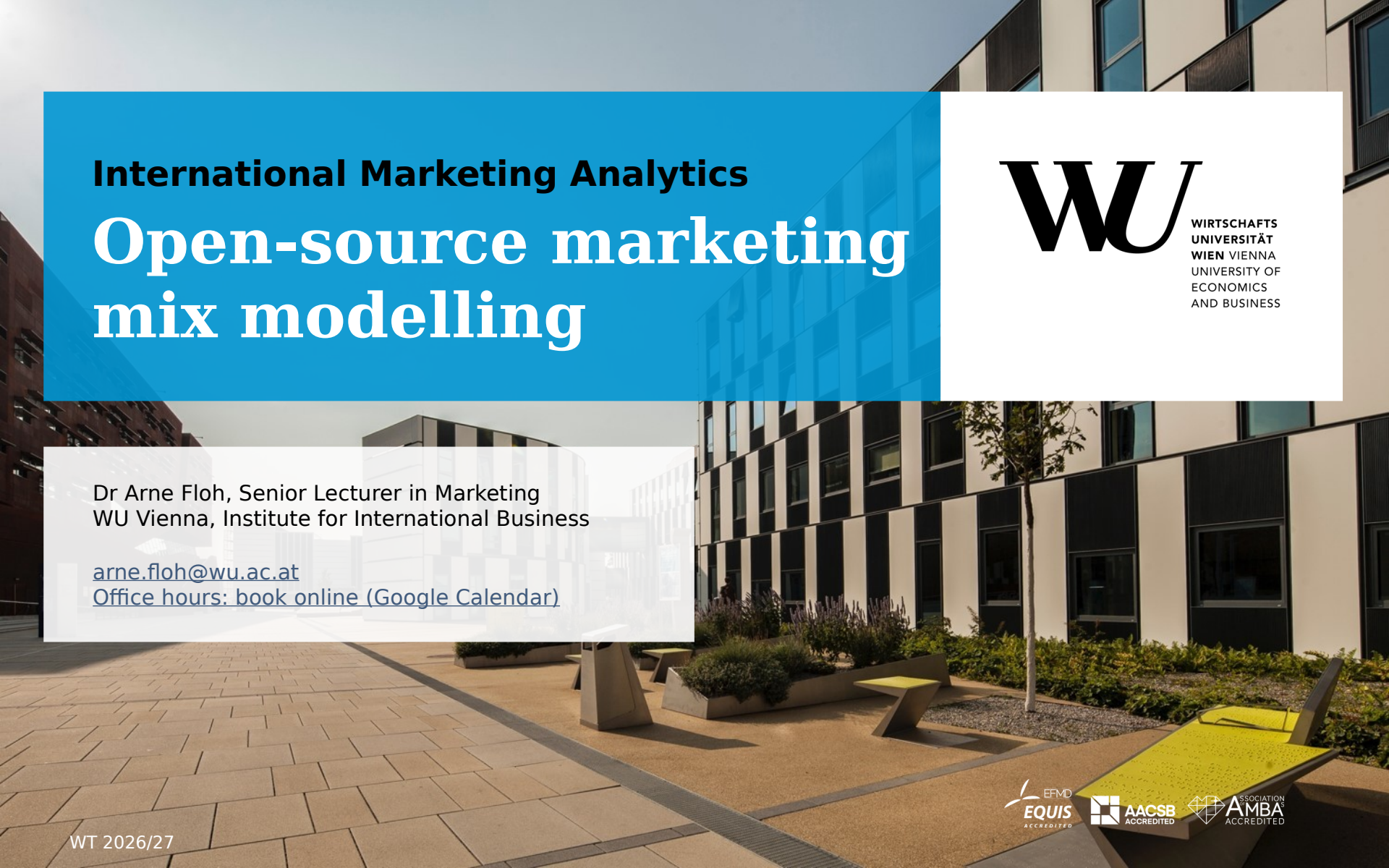


International Marketing Analytics

Open-source marketing mix modelling

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WIRTSCHAFTS UNIVERSITÄT WIEN VIENNA UNIVERSITY OF ECONOMICS AND BUSINESS

WT 2026/27

Today

Concepts → By hand → Three tools → Real data → Python → In class

Theory, Parts 1 to 3

- The model in one line, carryover, saturation, ROI
- A small example by hand
- PyMC-Marketing, Meridian, Robyn

Practice, Parts 4 to 6

- Real data: a sports web shop in Austria
- Python: a code ladder in nine steps
- Online quiz and an in-class assignment (no coding)

Question of the day: where should the next euro go, and how sure can we be?

Our example: where should the next euro go?

The problem: a brand spends on TV and online and must split next year's budget.

The question: what did each channel bring, and where should the next euro go?

Measure

sales from each channel

Explain

baseline vs media

Plan

move budget to the best next euro

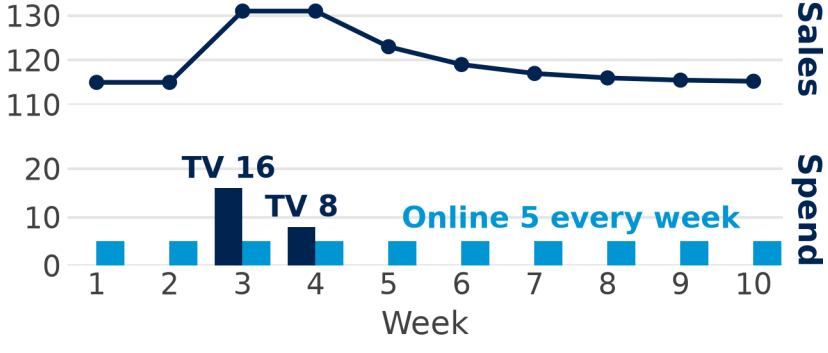
A tiny coffee brand first: 10 weeks, TV and online, by hand; a real web shop in Part 4.

Invented round numbers (hand_example.csv)

The data: ten weeks, one row per week

Coffee brand: spend and sales per week, k EUR

Week	1	2	3	4	5	6	7	8	9	10
TV	0	0	16	8	0	0	0	0	0	0
Online	5	5	5	5	5	5	5	5	5	5
Sales	**115**	**115**	**131**	**131**	**123**	**119**	**117**	**116**	**115.5**	**115.25**



Sales jump in the TV weeks 3 and 4: 131

and fade slowly: 123, 119, 117 …

Online: 5 k EUR every week

Invented round numbers (hand_example.csv)

The main concepts

Part 1 of 6 · the model in one line, carryover, saturation, ROI

Why MMM is back

Tracking breaks

Apple's opt-in rule and the GDPR hide many user journeys from the ad platforms.

No user data

Only weekly totals of sales and spend per market.

Free tools

PyMC-Marketing, Meridian and Robyn are open source.

Old idea, new tools: MMM dates back to the 1960s; Bayesian methods and free code made it affordable.

MMM works where tracking fails: it needs only weekly totals per market.

The model in one line

Sales = baseline + media effects + controls + noise

Baseline

what you would sell with no media: trend, season, holidays

Media effects

each channel after carryover and saturation

Controls

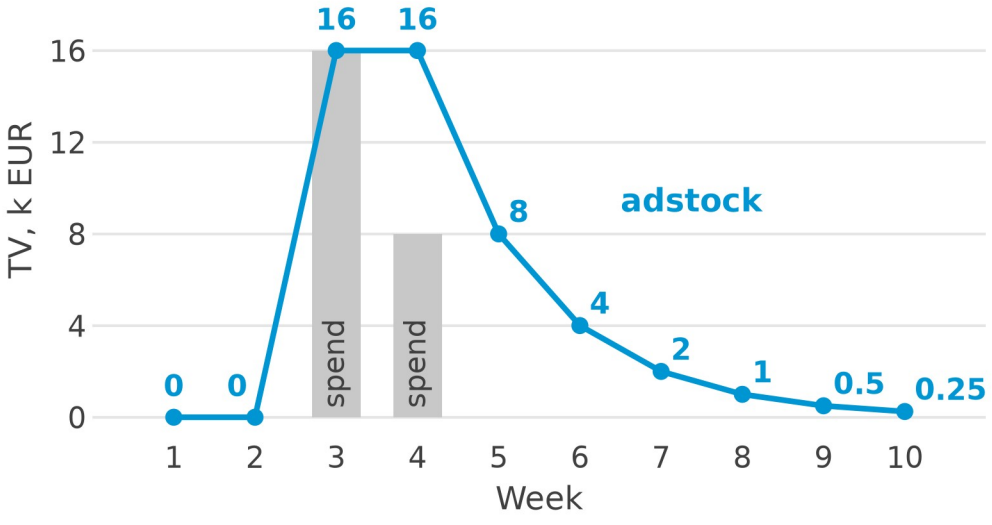
price, promotions, competitors, weather

Coffee brand, week 3: 100 + 1 × 16 + 3 × 5 = 131 — baseline 100 · TV adstock × 1 · online × 3

The output: the contribution and ROI of each channel, response curves, a budget plan.

Invented round numbers (hand_example.csv)

Carryover: this week's spend keeps working



Adstock = this week's spend + **decay** × last week's adstock

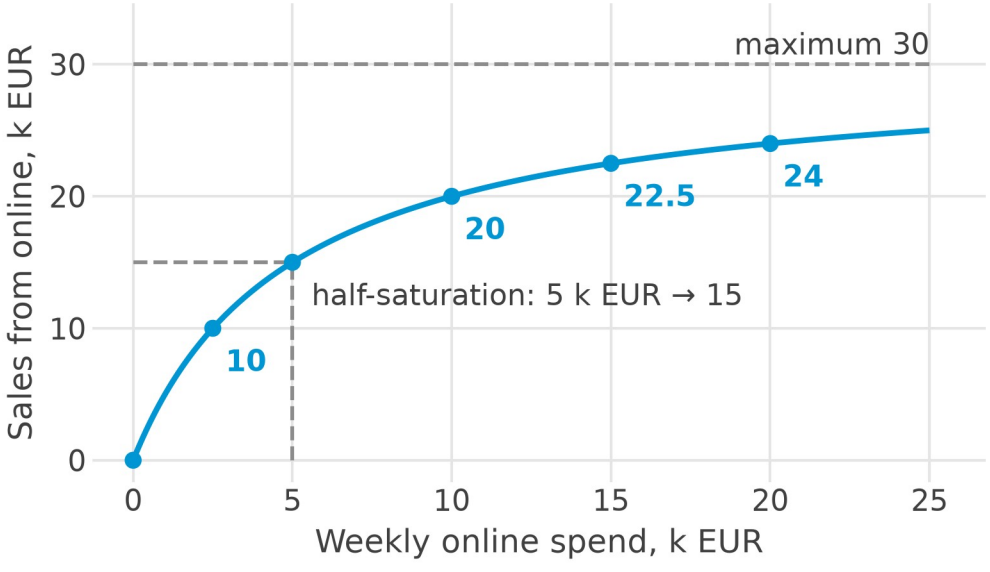
0.5 **decay:** half of last week's adstock is left this week

× 2 **each euro works twice:** 1 + 0.5 + 0.25 + 0.125 + … = 2

Carryover: TV in weeks 3–4 still sells in weeks 5–10.

Invented round numbers (hand_example.csv)

Saturation: each extra euro buys less



Online effect =
30 × spend / (spend + 5)

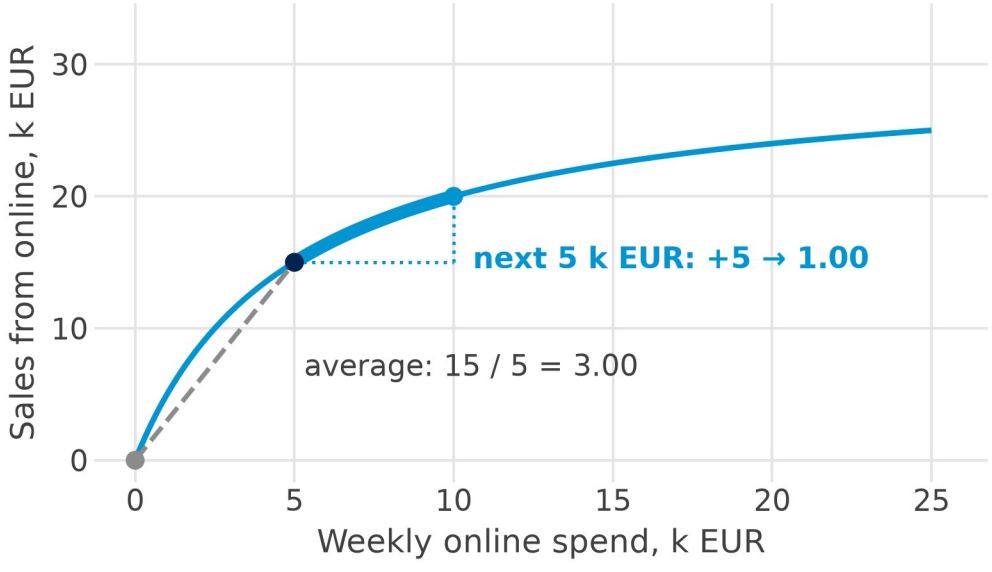
30 **maximum:** the most online can bring per week

5 **half-saturation:** at 5 k EUR online brings half of it, 15

Saturation: 10 k EUR bring 20, not twice 15.

Invented round numbers (hand_example.csv)

ROI and the response curve



3.00 **average ROI:** 15 / 5: 3 EUR of sales per euro

1.00 **next 5 k EUR:** (20 − 15) / 5 = 1.00

Response curve: weekly sales from a channel against its weekly spend

The shape of the curve, not the average ROI, decides where the next euro goes.

Invented round numbers (hand_example.csv)

A small example by hand

Part 2 of 6 · adstock, contributions, ROI, a budget question

Adstock week by week

Week	1	2	3	4	5	6	7	8	9	10
TV spend	0	0	16	**8**	0	0	0	0	0	0
+ 0.5 × last	0	0	0	**8**	8	4	2	1	0.5	0.25
= adstock	**0**	**0**	**16**	**16**	**8**	**4**	**2**	**1**	**0.5**	**0.25**

Week 4: 8 + 0.5 × 16 = 16
new spend + half of last week's adstock

24 TV spend in k EUR

Week 5: 0 + 0.5 × 16 = 8
no new spend: half is left

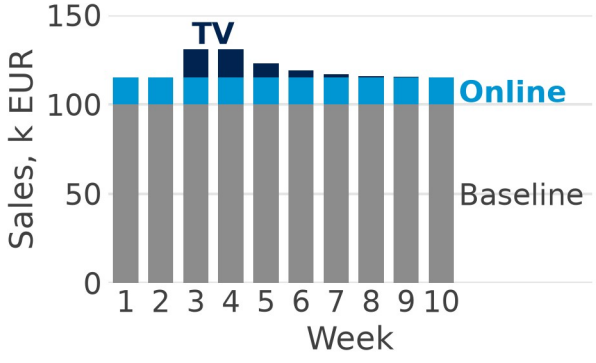
47.75 TV adstock weeks 1–10

Each euro works twice: 24 k EUR of TV spend count as 47.75 in ten weeks, 48 in the long run.

Invented round numbers (hand_example.csv)

Contributions and ROI by hand

Part	Calculation	Sales	Spend	ROI
Baseline	100 × 10 weeks	1,000	–	–
TV	1 × 47.75 adstock	47.75	24	1.99
Online	3 × 50 spend	150	50	3.00
Total		**1,197.75**	**74**	**–**



Sales per week: baseline, online, TV

ROI = sales ÷ spend. **Shares** of 1,197.75: baseline 83.5 %, online 12.5 %, TV 4.0 %

Ignore carryover and TV looks half as good: 24 / 24 = 1.00 instead of 1.99.

Invented round numbers (hand_example.csv)

The budget question: 5 k EUR more a week

The question: the brand has 5 k EUR more a week. TV or online?

TV

5 k EUR more TV work twice (carryover):

1 × 5 × 2 = 10 k EUR of sales

ROI of the next euro: 10 / 5 = 2.00

Online

Its curve (slide 9): 15 at 5, 20 at 10 k EUR:

20 − 15 = 5 k EUR of sales

ROI of the next euro: 5 / 5 = 1.00

Average ROI: online 3.00, TV 1.99: the average would pick the wrong channel.

Decide at the margin: put the next euro where the next euro sells most.

Invented round numbers (hand_example.csv)

Three open-source tools

Part 3 of 6 · PyMC-Marketing, Meridian, Robyn

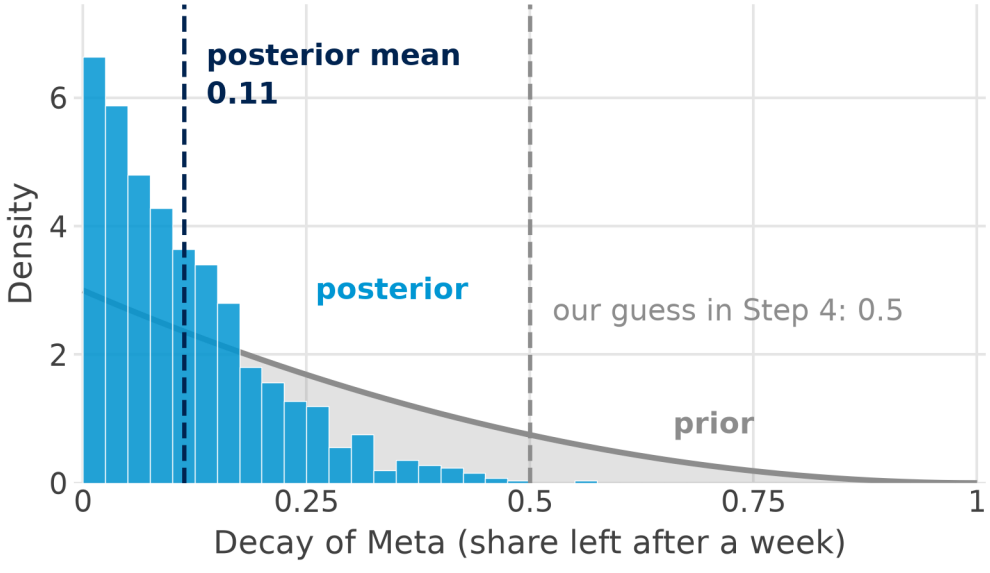
The three tools at a glance

	PyMC-Marketing	Google Meridian	Meta Robyn
Maintainer	PyMC Labs	Google	Meta Marketing Science
Version (Oct 2026)	1.3.0	2.1.0	3.12.1 (CRAN)
Language	Python	Python	R (Python port in beta)
Approach	Bayesian	Bayesian, geo-level	Ridge regression + search
Engine	PyMC	JAX	glmnet, Nevergrad, Prophet
Licence	Apache-2.0	Apache-2.0	MIT

All free and open, but two come from the largest ad platforms (slide 24).

Sources: PyPI, CRAN and the tools' repositories, checked 26 Oct 2026

Bayesian in one picture: prior and posterior



Prior: what we believe before the data (here: the decay is probably small, Beta(1, 3))

Posterior: the belief after the data: mean 0.11, 94 % interval 0.00 to 0.29

Check: R-hat 1.00, no divergences: the sampler agrees with itself

Bayesian MMM: every parameter gets a prior; the result is a range, not one number.

Prior: PyMC-Marketing default; posterior: the Austrian model of Step 8

PyMC-Marketing: Bayesian MMM in Python

What: PyMC Labs' free Python package (Apache-2.0), built on PyMC

How: a prior on every parameter; adstock, then saturation; every result comes with a range

Strengths: flexible; countries and brands in one model; lift tests and a budget optimiser built in

Best for: many markets or brands, a budget split across them, a team that codes in Python

open_source_mmm.qmd · Step 8

```
mmm = MMM(date_column="week", target_column="sales",
          channel_columns=["google", "meta"],
          control_columns=["week_no"],
          adstock=GeometricAdstock(l_max=6),
          saturation=LogisticSaturation(),
          yearly_seasonality=2)
```

After the fit you get	Slide
a posterior for every parameter	17, 39
contributions and ROI with a range	28
response curves, a budget plan	29

Our tool in Part 5: six lines define the model, one call fits it (Step 8).

Meridian: Google's Bayesian geo model

What: Google's free Python package (Apache-2.0); runs on JAX, a GPU helps

How: priors, as in PyMC-Marketing; regions (geos) in one model, each around a shared mean

Strengths: lift tests become ROI priors; reach and frequency for video; a guided Colab notebook

Best for: data by region and Google-heavy media

Not run here: not in the course environment

```
from meridian.data import (
    data_frame_input_data_builder as db)
from meridian.model import model, spec
builder = db.DataFrameInputDataBuilder(
    kpi_type="non_revenue",
    default_kpi_column="conversions")
builder = (builder.with_kpi(df)
           .with_population(df)
           .with_media(df,
               media_cols=["Channel0_impression"],
               media_spend_cols=["Channel0_spend"],
               media_channels=["Channel0"]))
mmm = model.Meridian(input_data=builder.build(),
                     model_spec=spec.ModelSpec())
mmm.sample_posterior(n_chains=10, n_adapt=1000,
                     n_burnin=500, n_keep=1000)
```

Geo first: Meridian needs population per region and models sales per head.

Source: Meridian Getting Started notebook, github.com/google/meridian (commit ef585b5); not run here

Robyn: Meta's ridge regression and search

What: Meta's free R package (MIT); a Python port is in beta

How: ridge regression (it shrinks the effects to keep them stable); a search tries thousands of adstock and saturation settings

Strengths: a quick first read-out; holidays for 123 countries built in; no sampler to check

Pick: the Pareto front keeps the models no other beats on fit and plausibility; the analyst picks one

Not run here: Robyn is an R package

```
library(Robyn)
InputCollect <- robyn_inputs(
  dt_input = dt_simulated_weekly,
  dt_holidays = dt_prophet_holidays,
  date_var = "DATE", dep_var = "revenue",
  dep_var_type = "revenue",
  prophet_vars = c("trend", "season", "holiday"),
  prophet_country = "DE",
  paid_media_spends = c("tv_S", "search_S"),
  adstock = "geometric",
  hyperparameters = hyperparameters)
OutputModels <- robyn_run(
  InputCollect = InputCollect,
  iterations = 2000, trials = 5)
```

Trade-off: priors make assumptions explicit; the Pareto front makes the analyst's choice explicit.

Source: Robyn demo.R, github.com/facebookexperimental/Robyn (commit aab4226); not run here

Calibrating with experiments

Lift test

Switch a channel on or off in some regions and measure the difference.

Example: more Meta in half of the regions, none in the rest.

Calibration: use that difference to anchor the model.

→ **Meridian:** the lift test becomes the ROI prior of that channel

→ **PyMC-Marketing:** each test is a point on the response curve, per country if needed

→ **Robyn:** tests are an extra objective; models far from the lift lose

International angle: tests often run only in large markets; a pooled Bayesian model passes that evidence on to the small ones.

Many countries: three ways to pool

No pooling



one model per country; small markets get noisy estimates

Robyn

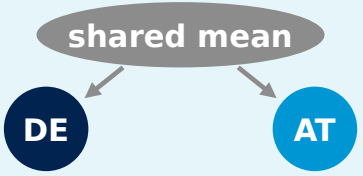
Full pooling



one effect for all countries; hides real differences

rarely sensible

Partial pooling



country effects around a shared mean; small markets borrow strength

Meridian, PyMC-Marketing

Our brand: Austria sold 3,180 k EUR in the same 134 weeks in which Germany sold 11,270

One currency first: Switzerland is in CHF; convert before you pool (ECB rates).

Data: Conjura MMM dataset (Anderson 2024), CC BY 4.0

Which tool for which situation?

PyMC-Marketing

- many countries or brands in one model
- a budget split across markets
- a team that codes in Python

Meridian

- geo-level data and Google-heavy media
- reach and frequency for video
- a guided Colab workflow

Robyn

- one market, few channels, R users
- a quick first read-out
- country holidays built in

Our shop: 2.6 years of weekly data in Austria; Germany next door

Rule of thumb: two or more years of weekly data per market; with less, pool countries or use informative priors.

Reading vendor claims: four questions

1. **Who wrote it?** Google, Meta or a partner may favour their own channels.
2. **Compared with what?** An uplift needs a baseline and a time window.
3. **Tested how?** A holdout period or a lift test, not only in-sample fit.
4. **Which version?** Results on old releases may no longer hold.

Free software is not a free model: the platforms that sell the media also wrote two of the tools that judge it.

Real data: a sports web shop

Part 4 of 6 · the data, a warning, the model, the next euro

The data: 134 weeks of a real web shop

File: sports_weekly.csv, Austria (AT)

week	sales	google	meta
26 Jul 2021	7.46	1.02	0.39
2 Aug 2021	9.32	1.36	0.27
9 Aug 2021	8.61	1.28	0.26
… 134 weeks in all			

sales: goods ordered in the web shop, k EUR

google: spend on all Google ads, k EUR

meta: spend on Facebook and Instagram ads, k EUR

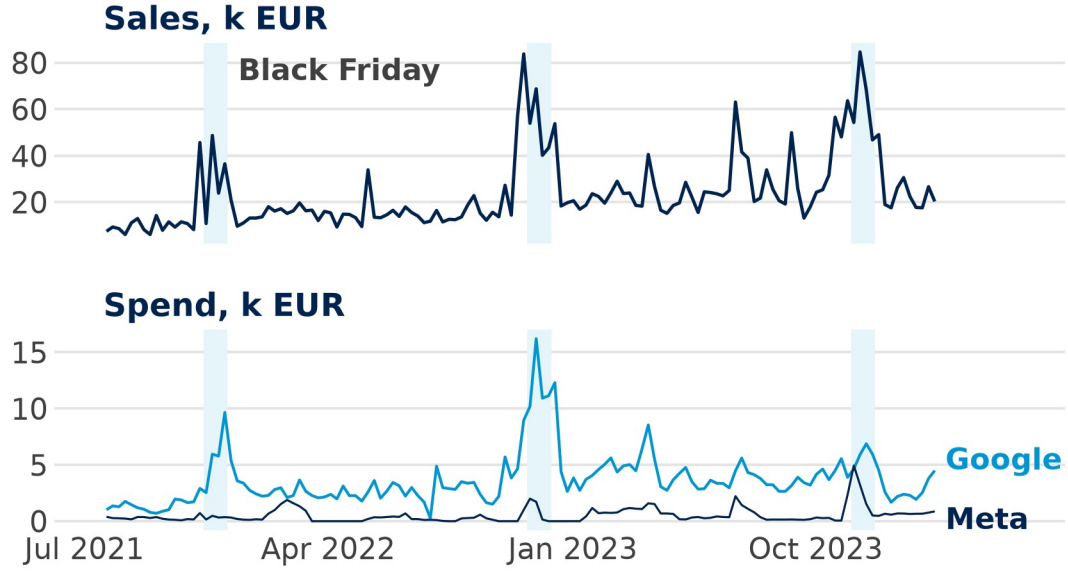
3,180 **k EUR of sales** in 134 weeks

561 **k EUR of spend:** Google 87.5 %, Meta 12.5 %

2.6 **years:** Germany and Switzerland are in the file too

Data: Anderson (2024), Conjura multi-region MMM dataset, figshare, CC BY 4.0; weekly sums by the course

Spend follows demand: a warning



7.03 **k EUR on Google** a week in the 10 best weeks ...

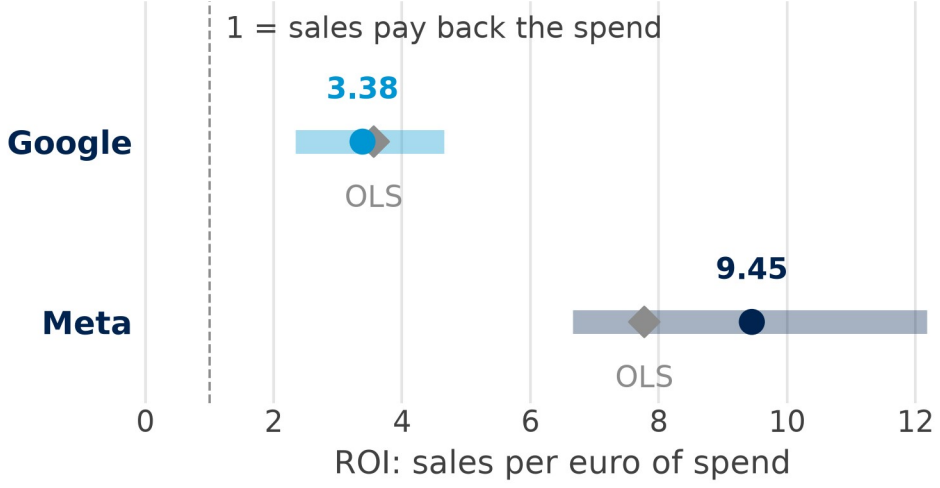
3.39 **... and** in the other 124 weeks

0.68 **correlation** of sales and Google spend

Spend follows demand: shopping ads spend more when more people search.

Data: Conjura MMM dataset (Anderson 2024), CC BY 4.0

What the model says: contributions and ROI



ROI with 94 % interval (PyMC-Marketing); ◆ OLS of Step 6

52 % **of sales from Google** (1,660 k EUR)

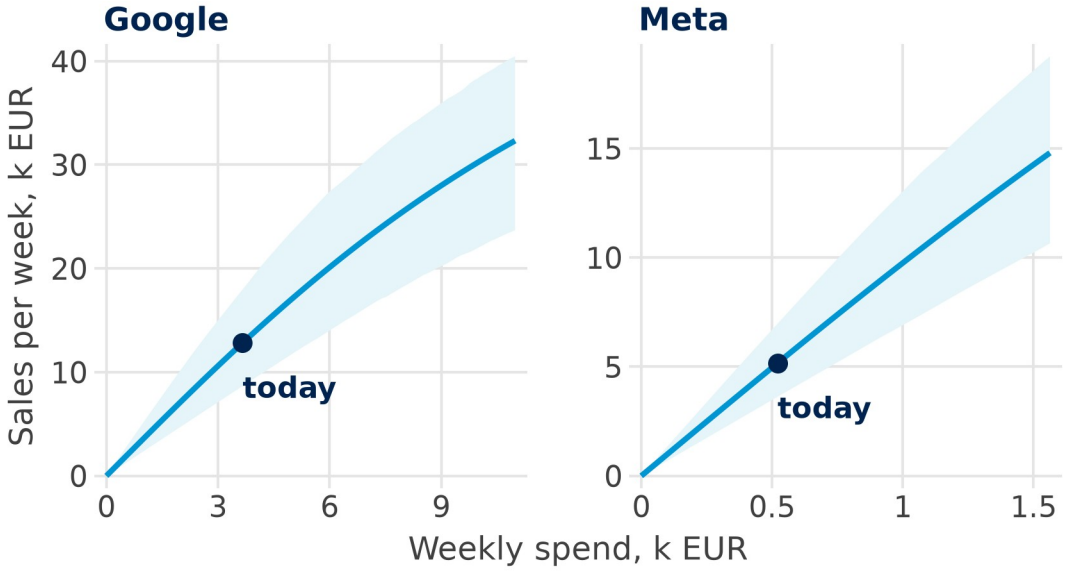
21 % **from Meta** (660 k EUR)

27 % **baseline,** trend and season

Meta: 9.45 per euro (6.66 to 12.18); Google: 3.38 (2.33 to 4.65).

Data: Conjura MMM dataset (Anderson 2024), CC BY 4.0; model: Step 8

Response curves: where the next euro works



9.77 **Meta:** next-euro ROI at today's 0.52 k EUR a week

3.28 **Google:** next-euro ROI at 3.66 k EUR a week

+3.1 **k EUR of sales a week** if 0.5 k EUR move from Google to Meta

Shift a little, then test: a geo lift test in Austria before the big move.

Data: Conjura MMM dataset (Anderson 2024), CC BY 4.0; curves: posterior of Step 8

MMM in Python



Part 5 of 6 · a code ladder, one step at a time

Step 1: load and look

open_source_mmm.qmd · Step 1

```
import polars as pl

# load the weekly data and keep Austria
file = "../../data/mmm_open/sports_weekly.csv"
df = pl.read_csv(file, try_parse_dates=True)
at = df.filter(pl.col("country") == "AT")
print(at.head())

# total sales and spend, k EUR
at.select(pl.col("sales", "google", "meta").sum())
```

week	sales	google	meta
26 Jul 2021	7.46	1.02	0.39
2 Aug 2021	9.32	1.36	0.27
9 Aug 2021	8.61	1.28	0.26
total	3,180	491	70

Look first: 134 weeks; Google gets 7 of every 8 euros.

Step 2: a first regression

open_source_mmm.qmd · Step 2

```
import statsmodels.formula.api as smf

# sales on the spend of both channels
m1 = smf.ols("sales ~ google + meta",
             data=at.to_pandas()).fit()
m1.params
```

term	coef
Intercept	5.14
google	3.93
meta	8.10
R^2	0.58

Before any control: a k EUR on Meta goes with 8.10 k EUR of sales.

Step 3: add the trend

open_source_mmm.qmd · Step 3

```
# week_no counts the weeks 1 to 134: the trend
at = at.with_columns(
    week_no=pl.int_range(1, pl.len() + 1))
m2 = smf.ols("sales ~ google + meta + week_no",
             data=at.to_pandas()).fit()
m2.params
```

term	coef
Intercept	0.57
google	3.60
meta	6.93
week_no	0.09
R^2	0.62

The trend takes its share: sales grew without the channels too.

Step 4: adstock in one line

open_source_mmm.qmd · Step 4

```
from scipy.signal import lfilter

# adstock: this week's spend + 0.5 x last stock
at = at.with_columns(
    meta_ad=lfilter([1], [1, -0.5], at["meta"]))
at.select("week", "meta", "meta_ad").head()
```

week	meta	meta_ad
26 Jul 2021	0.389	0.3890
2 Aug 2021	0.273	0.4675
9 Aug 2021	0.259	0.4928
16 Aug 2021	0.231	0.4774
23 Aug 2021	0.167	0.4057

As by hand: 0.273 + 0.5 × 0.389 = 0.4675 (slide 12).

Step 5: regression with carryover

open_source_mmm.qmd · Step 5

```
# Meta with its carryover, Google without
m3 = smf.ols("sales ~ google + meta_ad + week_no",
             data=at.to_pandas()).fit()
m3.params
```

term	coef
Intercept	0.58
google	3.56
meta_ad	3.93
week_no	0.09
R^2	0.60

Per k EUR of adstock: each euro of Meta counts about twice (Step 6).

Step 6: contributions and ROI

open_source_mmm.qmd · Step 6

```
# contribution = coefficient x column total
b = m3.params
c_google = b["google"] * at["google"].sum()
c_meta = b["meta_ad"] * at["meta_ad"].sum()
print(c_google, c_meta)

# ROI = contribution / spend
print(c_google / at["google"].sum())
print(c_meta / at["meta"].sum())
```

channel	contrib.	ROI
google	1,746	3.56
meta	543	7.78
baseline	891	–



Meta returns more per euro, Google brings more sales.

Step 7: saturation in one line

open_source_mmm.qmd · Step 7

```
# saturation: half-saturation at 20 k EUR a week
at = at.with_columns(
    google_sat=at["google"] / (at["google"] + 20))
m4 = smf.ols(
    "sales ~ google_sat + meta_ad + week_no",
    data=at.to_pandas()).fit()
print(m3.rsquared, m4.rsquared)

# the Google ROI of the curved model
b4 = m4.params
c_sat = b4["google_sat"] * at["google_sat"].sum()
print(c_sat / at["google"].sum())
```

model	R^2	Google ROI
straight	0.60	3.56
curved	0.60	4.79

Same fit, other answer: the data cannot choose the curve.

Step 8: PyMC-Marketing

open_source_mmm.qmd · Step 8

```
from pymc_marketing.mmm import (
    GeometricAdstock, LogisticSaturation)
from pymc_marketing.mmm.multidimensional import MMM

# the Bayesian MMM: adstock, then saturation
mmm = MMM(date_column="week", target_column="sales",
          channel_columns=["google", "meta"],
          control_columns=["week_no"],
          adstock=GeometricAdstock(l_max=6),
          saturation=LogisticSaturation(),
          yearly_seasonality=2)
X = at.select("week", "google", "meta", "week_no")
mmm.fit(X.to_pandas(), at["sales"].to_pandas(),
        draws=500, tune=500, chains=2,
        random_seed=1, progressbar=False)
```

setting	value
chains × draws	2 × 500
adstock window	6 weeks
season	yearly, 2 waves
runtime	about 25 s

One model, all at once: adstock, saturation, trend and season, with priors.

Step 9: check and read

open_source_mmm.qmd · Step 9

```
import arviz as az

# the decay of each channel, with R-hat
print(az.summary(mmm.idata,
                 var_names=["adstock_alpha"]))

# ROI per channel: mean and 94 % interval
roi = mmm.summary.roas(frequency="all_time")
roi
```

result	google	meta
decay	0.05	0.11
R-hat	1.00	1.00
ROI	3.38	9.45
94 % interval	2.33–4.65	6.66–12.18

Same ranking as OLS, now with a range: and Meta's decay is 0.11, not 0.5.

In-class assignment

Part 6 of 6 · quiz, four tasks, no coding

Online quiz: MMM in five questions

1 Decay 0.5, last week's adstock 6, spend this week 10. This week's adstock is …
A 16 B 13 C 8

2 A channel brought 120 k EUR of sales for 40 k EUR of spend. Its ROI is …
A 0.33 B 3.00 C 80

3 Sales 30 at 10 k EUR a week, 36 at 20 k EUR a week. The ROI of the extra 10 k EUR is …
A 3.00 B 1.80 C 0.60

4 Which tool fits many ridge regressions and lets the analyst pick from a Pareto front?
A PyMC-Marketing B Meridian C Robyn

5 A new market has six months of weekly data. The best start is …
A a model of its own B partial pooling with the other markets C the biggest market's ROI

3 minutes alone, then compare with your neighbour.

Answers: in class

Your turn: an MMM review in four tasks

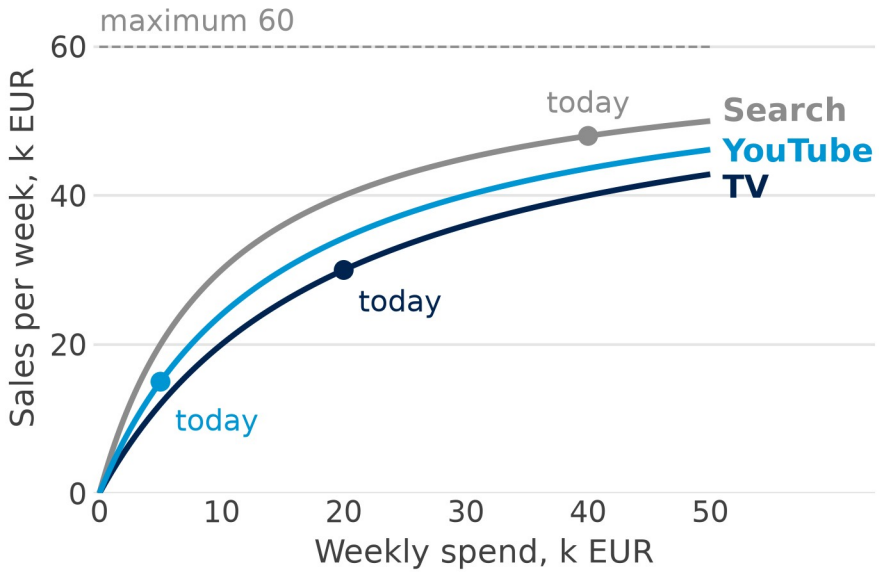
Groups of three: 35 minutes	Plenary: 10 minutes

1. **Budget:** three response curves of an e-bike brand: where should 10 k EUR a week go? (slide 44)
2. **Model check:** the German model credits Google and Meta with 95 % of sales. Plausible? What would you test?
3. **Vendor claim:** grade one case study with the four questions (slide 24)
4. **Tool choice:** PyMC-Marketing, Meridian or Robyn for two brands, with two reasons each (slide 23)

Deliverable: one slide per group: the budget move, the verdict on Germany, the grade, the two tools.

Handout: `sessions/02-mmm-one-country/mmm_assignment.qmd` (no code needed)

Task 1: three response curves



Now	Spend	Sales	Avg ROI
TV	20	30	1.50
YouTube	5	15	3.00
Search	40	48	1.20

Sales at	−10 k EUR	+10 k EUR
TV	20	36
YouTube	–	30
Search	45	50

Your move: which channel gives 10 k EUR a week, which one gets it, and why?

Invented round numbers (assignment)

Open data for MMM projects

Conjura eCommerce MMM

Real, anonymised daily data of about 100 web shops; Google, Meta, TikTok

Licence: CC BY 4.0 (our Part 4)

Meridian geo data

Simulated weekly panels, many regions, reach and frequency

Use: geo vs national model

PyMC-Marketing data

A two-region panel with true parameters, lift-test data

Use: can the model find the truth?

Robyn demo data

Simulated weekly spend and revenue; holidays of 123 countries

Use: an R lab, holiday effects

The licence decides: commit only data you may share; link the rest.

Country controls: Eurostat and World Bank (prices, population, internet use)

Sources: dataset pages; links in open_source_mmm.qmd (appendix)

Key takeaways

1. **One line:** sales = baseline + media effects + controls + noise
2. **Carryover and saturation:** spend keeps working, and each extra euro buys less
3. **Decide at the margin:** the next euro, not the average ROI
4. **Three tools:** two Bayesian tools that pool markets, one ridge tool per market
5. **Be sceptical:** spend follows demand; calibrate with experiments; ask who wrote the case

Next: run sessions/02-mmm-one-country/open_source_mmm.qmd; it reproduces Parts 4 and 5.

References

MMM methods

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Gelman et al. (2020). Bayesian Workflow, arXiv 2011.01808.

Tools and data

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Meridian (Google). Apache-2.0, github.com/google/meridian.

Robyn (Meta Marketing Science). MIT, github.com/facebookexperimental/Robyn.

Anderson, A. (2024). Multi-Region Marketing Mix Modelling (MMM) Dataset for Several eCommerce Brands (v3). figshare. doi.org/10.6084/m9.figshare.25314841.v3. CC BY 4.0.

Free to read: MSI Working Paper 26-106 (Runge & Pauwels), arXiv 2011.01808 (Bayesian workflow)

Thank you. Questions and discussion

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